Introduction

The objective of this project was to analyse the performance of a social media advertising campaign using real-world marketing data and present insights through an interactive Power BI dashboard.

The goal was not only to calculate metrics, but to understand:

- how well the campaign performed overall,
- which campaigns (posts/ads) delivered the best results,
- how efficiently money was spent,
- and what actions can improve future campaigns.

Throughout the analysis, special attention was given to data accuracy, meaningful visualisation, and practical business interpretation.

Overview of Campaign KPIs

What I did

I started by creating a high-level KPI overview using card visuals in Power BI. These KPIs give a quick snapshot of campaign performance and are usually the first thing stakeholders look at.

I calculated and displayed the following metrics:

- Total Impressions
- Total Clicks
- Total Conversions
- Click-Through Rate (CTR)
- Cost Per Click (CPC)
- Return on Investment (ROI)

Some of these metrics (CTR, CPC, ROI) were created using custom DAX measures to ensure correct calculations.

Why I did it

KPIs help answer the most basic but important question:

> "Did the campaign perform well overall?"

They allow decision-makers to understand campaign health without going into details.

Key results

79M
Total Impressions

13K
Total Clicks

585
Total Conversions

1.69%
CTR

$1.51
CPC

135.42%
ROI

These values show that the campaign achieved strong reach and generated a positive return, meaning the campaign earned more value than it cost.

Insights into Top-Performing Posts (Campaigns)

What I did

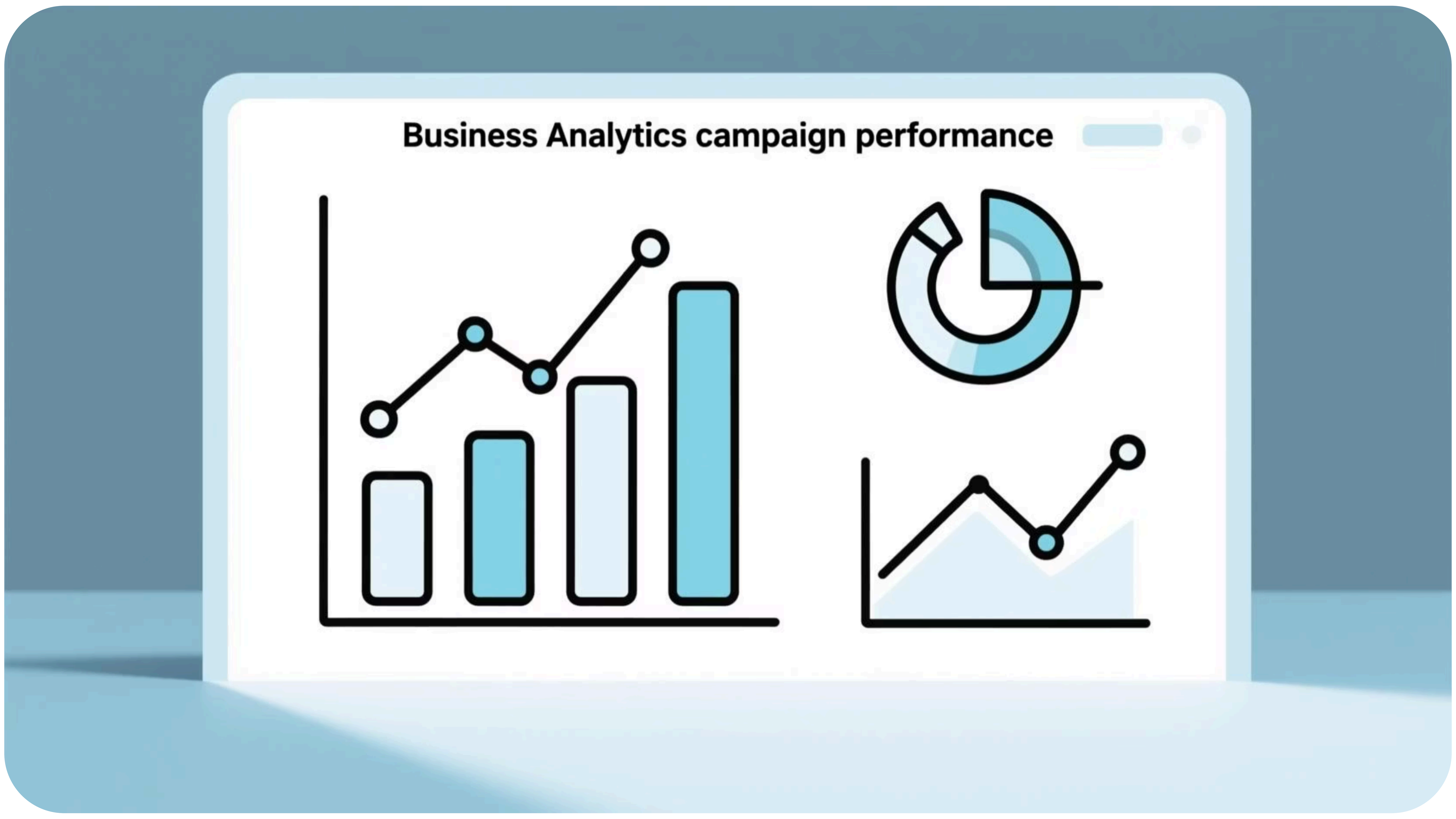
To understand performance at a deeper level, I analysed results by campaign (which represents individual ads or posts).

I created:

- Clicks by Campaign chart
- CTR (%) by Campaign chart
- ROI (%) by Campaign chart

These charts help compare campaigns from different angles:

- volume (clicks),
- engagement quality (CTR),
- profitability (ROI).



Why I did it

Looking only at overall KPIs can hide important differences. Some campaigns may get many clicks but low returns, while others may be smaller but highly efficient.

Campaign-level analysis answers:

> "Which campaigns should we scale, optimise, or stop?"

Key insights

Campaign 1178
generated the highest number of clicks and conversions, showing strong reach and audience interest.

Campaign 936
showed stable engagement and moderate ROI, making it suitable for optimisation.

Campaign 916
had lower volume but the highest ROI, indicating strong efficiency.

This shows that the campaign with the most clicks is not always the most profitable.

ROI Summary

What I did

I calculated Return on Investment (ROI) using a DAX formula that compares conversion value against total spend. ROI was analysed at:

- an overall campaign level (KPI card),
- an individual campaign level (bar chart).

Why I did it

ROI is one of the most important metrics for businesses because it directly answers:

> "Are we making money from this campaign?"

High engagement without profitability does not justify continued spending.

Key findings

The campaign achieved an overall ROI of 135.42%, indicating strong profitability.

Campaign 916 delivered the best ROI despite lower scale.

Campaign 1178, while driving volume, showed lower ROI due to higher spend.

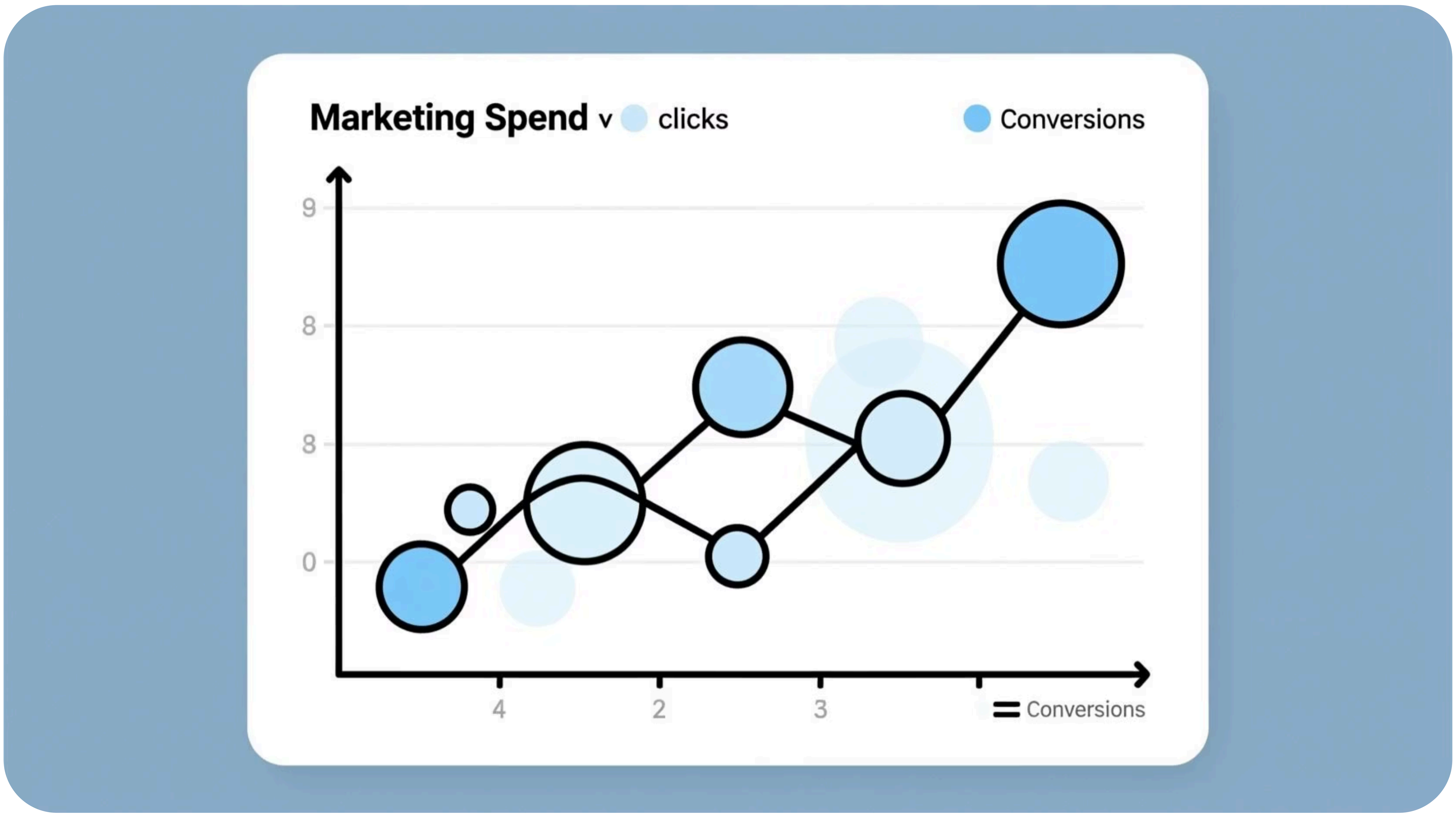
This analysis highlights the importance of balancing scale and efficiency.

Conversion and Cost Efficiency Analysis

What I did

To move beyond clicks, I analysed:

- Conversions by Campaign
- Cost per Conversion by Campaign
- Spend vs Clicks scatter chart, where:
    - X-axis = Spend
    - Y-axis = Clicks
    - Bubble size = Conversions



Why I did it

Clicks alone do not represent business success. Conversions and cost efficiency show whether ad spend is being used wisely.

The scatter chart helps identify:

- high-spend but low-return campaigns,
- low-spend but high-performing opportunities.

Insights

Some campaigns convert users at a much lower cost than others.

High-spend campaigns are not always the most efficient.

Cost per conversion is a better optimisation metric than clicks alone.

Audience Insights and Data Decisions

What I did

I analysed audience engagement using gender-based insights.

While exploring the dataset, I noticed that the age column contained inconsistent and unreliable values, which could lead to misleading insights.

Why I did it

Instead of forcing incorrect data into visuals, I chose to:

- exclude unreliable age analysis, and
- focus only on clean and interpretable audience data.

This ensures the dashboard remains accurate and trustworthy.

Key insight

Male users accounted for the majority of clicks, suggesting potential for targeted optimization.

Actionable Recommendations

Based on the analysis, the following actions are recommended:

01

Increase investment in high-ROI campaigns

Campaigns that generate strong ROI should be scaled gradually while monitoring efficiency.

02

Optimise underperforming campaigns

Campaigns with low CTR or high cost per conversion should be improved through better creatives or targeting.

03

Focus on cost per conversion, not just clicks

Future optimisation should prioritise conversions and ROI over raw engagement numbers.

04

Use audience insights for targeting

Gender-based engagement trends can be used to refine messaging and targeting strategies.

05

Maintain data quality in reporting

Only reliable and meaningful data should be visualised to support accurate decision-making.

Conclusion

This project demonstrates a complete workflow of campaign performance analysis using Power BI - from data preparation and KPI calculation to campaign comparison and actionable insights.

The final dashboard provides a **clear**, **interactive**, and **business-focused** view of marketing performance, enabling informed decisions for future campaigns.